

Grade 7
Unit 3
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

1. Delaney purchased a new boat for \$61,014.00. She read that the value of new boats change by $\frac{-2}{19}$ in the first year. What is the value of Delaney's boat after the first year?

$$61,014 \left(\frac{-2}{19} \right) = -6,422.53$$
$$\begin{array}{r} 61014.00 \\ - 6422.53 \\ \hline \$54,591.47 \end{array}$$

2. Delaney learns that the value of her boat will continue to change by $\frac{-3}{10}$ of its value each year after the first year. Using the value after year one from Question 1, what is the value of Delaney's boat at the end of year 2?

$$54,591.47 \left(\frac{-3}{10} \right) = -16,377.44$$
$$\begin{array}{r} 54591.47 \\ - 16377.44 \\ \hline \$38,214.03 \end{array}$$

3. An Orca whale is swimming $-176\frac{1}{4}$ ft below sea level when he notices the rest of his pod above him at the surface. Orca whales can swim at a rate of 120.52 feet per hour. How long, in hours, will it take the Orca to reach his pod?

$$\frac{176.25}{120.52} \approx 1.46$$

Approximatley 1.46 hours

4. A candle store sells the 3-wick candles for $2\frac{1}{5}$ times what they paid for them from the warehouse. The candles are purchased from the warehouse at \$5.65. How much profit does the candle store make off each candle?

$$2.2(5.65) = 12.43$$
$$\begin{array}{r} 12.43 \\ - 5.65 \\ \hline \$6.78 \end{array}$$

5. After 3 months, the price of the candles is reduced by $\frac{1}{5}$ of the original store price. Using the sale price from Question 4, what is the price of the candles after the 3 month reduction?

$$12.43(0.2) = 2.49$$
$$\begin{array}{r} 12.43 \\ - 2.49 \\ \hline \$9.94 \end{array}$$

6. Marie's office has a parking garage that charges \$10.50 each day for parking. Marie's employer provides her with a parking card that has \$200 each month. When Marie parked today, the card indicated it had \$116 left on the card. How many days has Marie used the card for parking this month?

$$\frac{(200-116)}{10.50} = 8 \text{ days}$$

7. Tanner is traveling on his first flight to visit family. $8\frac{4}{5}$ minutes into the flight, the pilot announces the plane is currently at an altitude of 3740 feet. At that rate, how long will it take to reach a cruising altitude of 32,400 feet?

$$\frac{8.8}{3740} = \frac{x}{32400} \quad \frac{285,120}{3,740} \approx 76.2 \text{ minutes}$$

8. A penguin sitting on an iceberg notices a fish swimming below. The fish is 10.3 ft below sea level. The penguin dives and reaches the fish in $\frac{7}{15}$ of a minute. What was the rate of descent the penguin made?

$$\frac{10.3 \text{ ft}}{\frac{7}{15} \text{ min}} \approx 22.07 \text{ feet per minute}$$

9. The temperature in North Dakota today was -8.8°C . Over the next 3 weeks, the temperature will increase an average of 3.2°C each week. What will the temperature be at the end of the 3 weeks?

$$3.2(3) = 9.6 \quad -8.8 + 9.6 = 0.8^{\circ}\text{C}$$

10. Lucy buys her cats a bag of food that is 35.3 ounces. Each of Lucy's cats eat $\frac{9}{15}$ ounces a day. If she has 2 cats, how long will the bag of cat food last Lucy?

$$\frac{35.3}{\left(2 \times \frac{9}{15}\right)} = 29\frac{5}{12} \text{ days or approximately 29 days}$$